

NeurIPS 2022 | Unfolding a Detector One Moment at a Time

Moment Unfolding recovers distribution moments directly from detector data, without ever binning, using a GAN-inspired reweighting whose knobs are the moments themselves.

Every measurement a collider records is blurred. A detector never sees the underlying physics cleanly; it smears energies, misplaces angles, and reshapes the distributions physicists want to study. To compare an experiment with theory, that blur has to be removed. The field calls this step unfolding, or deconvolution, one of the quiet workhorses of particle and nuclear physics.

The standard recipe has an awkward habit: it first pours the data into a histogram. You bin the observable, unfold the whole binned spectrum, and only then read off the summary you actually cared about. But binning is a lossy convenience. It introduces discretization artifacts, and it is wasteful when what you truly want is not the entire spectrum but a handful of statistical moments, such as how a jet's width grows with energy. Many theory predictions sit at exactly that level: perturbative QCD cannot hand you the full density of a hadronic jet, yet it can predict the energy dependence of its moments.

A new NeurIPS 2022 paper from Krish Desai (UC Berkeley), Benjamin Nachman (Lawrence Berkeley National Laboratory), and Jesse Thaler (MIT) closes that gap directly. Their method, Moment Unfolding, skips the histogram entirely and unfolds the moments of an observable as first-class quantities. It is unbinned, non-iterative, and borrows its machinery from Generative Adversarial Networks (GANs) and, more surprisingly, from Boltzmann's nineteenth-century statistical mechanics.

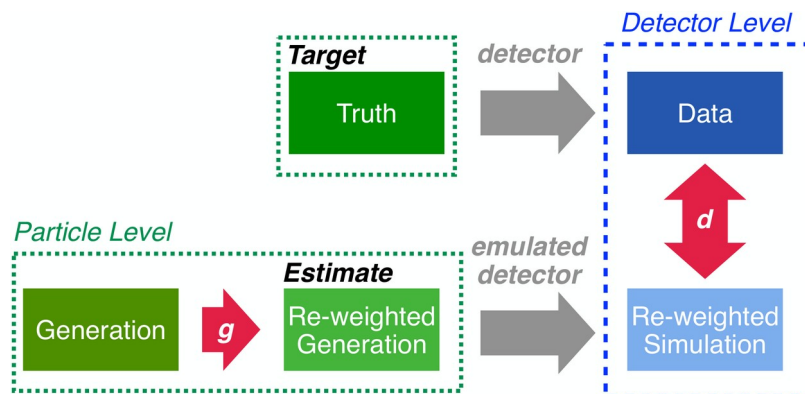


Figure 1. The Moment Unfolding setup. A generator g reweights the particle-level generation so that, after the same detector emulation, the reweighted simulation becomes statistically indistinguishable from the detector-level data. A discriminator d is charged with telling the two apart. Because the reweighting changes only importance weights and not the events themselves, the expensive detector emulation is run once.

Paper: https://ml4physicsciences.github.io/2022/files/NeurIPS_ML4PS_2022_43.pdf

Code: <https://github.com/hep-lbdl/MomentUnfolding>

The binning tax

To see why moments deserve their own method, it helps to picture the usual two-step dance. Suppose you want the moments of an observable X as a function of another quantity Y , for instance a jet's width as a function of its momentum. Conventional tools discretize the joint (X, Y) support into a two-dimensional histogram, unfold that histogram, and compute moments of X within each bin of Y . Every bin edge is a small modeling choice, and every such choice stamps a discretization footprint onto the very numbers a physicist ultimately reports to the rest of the field.

Unbinned unfolding methods already exist and, by construction, avoid these artifacts. But they are generalists: they aim to reconstruct an entire spectrum, so they can trade away precision on any one aspect of it, including the small set of moments a physicist may actually be after. Moment Unfolding takes the opposite stance. Instead of unfolding everything and summarizing afterward, it targets the chosen moments from the start and lets the rest of the distribution stay unconstrained.

A generator whose knobs are the moments

The clever move is in how the generator is parametrized. Boltzmann built the distribution that maximizes entropy subject to a fixed mean energy, and the result is an exponential of the conserved quantity. Moment Unfolding reuses that shape. Its generator is written as $g(x) = \exp(\lambda_1 x + \lambda_2 x^2 + \dots + \lambda_n x^n)$, the exponential of a degree- n polynomial. This is a deliberately tiny neural network with only n trainable parameters, the coefficients λ_i , and those coefficients map directly onto the moments being unfolded. Turn the knobs, and you are literally adjusting the moments.

That generator sits inside a GAN, an adversarial training contest. It reweights the simulated events; a discriminator d , a small network with three hidden layers of 50 ReLU nodes and a sigmoid output, tries to distinguish the reweighted simulation from the real data. The two are trained against each other on a single weighted binary cross-entropy loss: the discriminator minimizes it while the generator maximizes it. When the discriminator can no longer tell the reweighted simulation from the data, the generator's coefficients are the unfolded moments.

One design detail matters for cost and for correctness. Because reweighting alters importance weights rather than the event features, the detector emulation does not need to be rerun as the weights change. It runs a single time. This is also where the method parts ways with an iterative reweighting approach such as OmniFold, which trains a fresh pair of networks on every iteration. Moment Unfolding trains one pair of networks, once.

A Gaussian sanity check

Before touching real physics, the authors run a controlled toy problem where the right answer is known in advance. The truth is a standard normal $N(0,1)$; the generation is shifted to $N(-0.5,1)$; and the detector is emulated as additive Gaussian noise $N(0,5)$. With 10^6 samples split 3:1 for training and testing, and because a Gaussian is fully described by finitely many moments, unfolding its moments here is equivalent to unfolding the entire density.

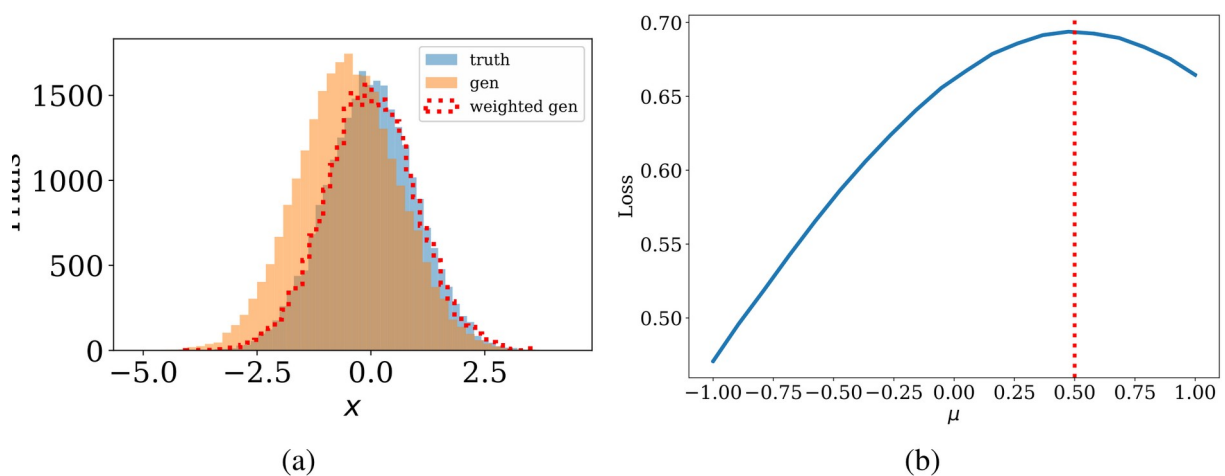


Figure 2. Gaussian sanity check. (a) Histograms of truth, generation, and the reweighted generation. (b) Scanning the loss over a candidate mean produces a curve whose peak sits exactly at the true value (red dotted line), confirming the procedure recovers the right answer.

The reweighted generation lands on top of the truth, and the diagnostic in panel (b) makes the success legible: scanning the loss as a function of the candidate mean gives a curve that peaks precisely at the true mean. Training is short, with the discriminator converging well within 10 epochs.

Real jets, two moments at once

The real test uses hadronic jets from simulated LHC collisions, the same Pythia and Herwig plus Delphes datasets introduced with OmniFold. One simulation plays the role of data and the other of the synthetic reference, so the truth is known and the answer can be graded. The observable is the jet width, a one-dimensional substructure variable whose distribution rises and then falls off rapidly past roughly a tenth. Here the method unfolds two moments of the jet width simultaneously, the mean and the second moment.

Reading the two panels of the next figure together is the whole point. The left panel overlays the particle-level distributions of truth, generation, and reweighted generation, so the quality of the match is visible by eye; the right panel turns that agreement into a number by scanning the loss over candidate values of each moment. Because the jet width is bounded and sharply peaked, the low-width region carries most of the statistical weight, and that is exactly where the reweighting has to work hardest to line the reweighted generation up with the truth.

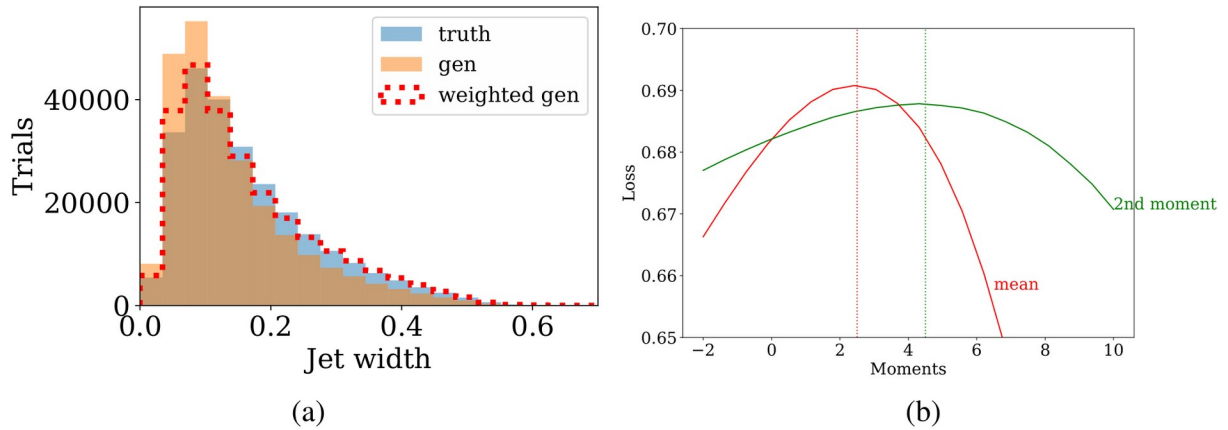


Figure 3. Jet width from simulated LHC collisions. (a) Particle-level histograms of truth, generation, and reweighted generation. (b) Scanning the loss over each candidate moment yields curves whose peaks land on the known true moments (vertical dotted lines), with a mean absolute error of $\leq 0.02\%$.

Panel (b) is the payoff. Each curve is a slice of the loss taken at the true value of the other moment, and the peaks of the two curves fall on the vertical dotted lines that mark the moments known before training. Predicted and true moments agree with a mean absolute error of $\leq 0.02\%$, comfortably sub-percent.

The paper is candid about the flip side. Even after the first and second moments of the reweighted generation match the truth, the full jet-width distributions are still not statistically identical. That is not a bug; it is the design. Higher moments differ between truth and generation and were never part of the fit, so they remain free. Moment Unfolding controls exactly the moments you ask for and leaves the rest alone, which is precisely the scoping that a moment-level theory comparison wants.

The numbers

Table 1. Headline results for Moment Unfolding.

Setting	Value
Moment recovery error (jet width)	$\leq 0.02\%$ MAE
Moments unfolded simultaneously	2 (mean and second)
Discriminator convergence	within 10 epochs
Gaussian study samples	10^6 , split 3:1 train/test
Discriminator network	3 hidden layers, 50 ReLU nodes
Full reproduction	under 5 minutes on a single Nvidia RTX6000 GPU

Why it matters, and where it stops

Moment Unfolding is a compact idea with a clean statement: make your generator's parameters the very quantities you want to measure, then let a discriminator drive them to the truth. It recovers moments to better than a hundredth of a percent on realistic LHC jet simulations, trains only once, and never bins the data. The whole analysis reproduces in under 5 minutes on a single GPU, lowering the barrier for others to build on it.

The honest boundary is that this extended abstract handles a small, chosen set of moments in one dimension; the authors flag higher-dimensional cases and comparisons with other unfolding methods as future work, and pushing to infinitely many moments runs into open partition-function questions. Still, the reweighting algorithm is agnostic to the dataset. Deconvolution shows up far beyond colliders, from medical devices and seismology to spectral astronomy and computer vision, and the same moment-level trick could travel with it.

For a practitioner, the appeal is modularity. You decide which moments matter for the comparison you want to make, wire them into the generator as its parameters, and let the adversarial game do the fitting; the discriminator never needs to know that its opponent is really a moment estimator in disguise. Nothing in that loop is specific to jets or even to physics, which is what makes the idea worth watching as unbinned unfolding matures and as similar deconvolution problems surface in other fields.